

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2301

Repetto 138 kV
March 30, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2301. The Interconnection Customer proposes a 146.2 MW solar photovoltaic facility interconnecting at the Repetto 138 kV substation in the Virginia area. The request was received on March 17, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 696 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,889,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SALEM ELECTRIC DEPARTMENT SUBSTATION / REPETTO 138 KV
Contingency: REPETTO / ASH STREET 138 KV
Loading: 155.4% (Rating: 211 MVA)
DFAX: 100.0%
Pre-project loading: 86.1%
2. Facility: NEOSHO / SALEM ELECTRIC DEPARTMENT 138 KV
Contingency: ASH STREET / REPETTO 138 KV
Loading: 143.1% (Rating: 201 MVA)
DFAX: 100.0%
Pre-project loading: 70.5%
3. Facility: MICHIGAN POWER LP / BLUE RIDGE 138 KV L
Contingency: REUSENS / HUNTINGTON COURT 345 KV
Loading: 138.2% (Rating: 422 MVA)
DFAX: 32.3%
Pre-project loading: 127.0%
4. Facility: BOONSBORO / REUSENS 138 KV
Contingency: REUSENS / LEESVILLE 345 KV
Loading: 105.9% (Rating: 662 MVA)
DFAX: 21.9%
Pre-project loading: 101.1%
5. Facility: TRAIN / RALLY HILL 138 KV Line
Contingency: REUSENS / HUNTINGTON COURT 345 KV
Loading: 103.6% (Rating: 315 MVA)
DFAX: 19.1%
Pre-project loading: 94.7%

6. Facility: STEAMBOAT / HUNTINGTON COURT 138 KV line
Contingency: HUNTINGTON COURT / HANCOCK 345 KV
Loading: 102.5% (Rating: 435 MVA)
DFAX: 25.8%
Pre-project loading: 93.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SALEM ELECTRIC DEPARTMENT SUBSTATION / REPETTO 138 KV	Rebuild line with higher-capacity conductor	\$2,323,000	42% (\$988,000)
NEOSHO / SALEM ELECTRIC DEPARTMENT 138 KV	Rebuild line with higher-capacity conductor	\$848,000	42% (\$360,000)
BLUE RIDGE / MICHIGAN POWER LP 138 KV	Rebuild line with higher-capacity conductor	\$5,698,000	5% (\$295,000)
BOONSBORO / REUSENS 138 KV	Reconductor line	\$1,346,000	3% (\$45,000)
RALLYHILL / TRAIN 138 KV line	Reconductor line	\$2,039,000	6% (\$121,000)
STEAMBOAT / HUNTINGTON COURT 138 KV	Reconductor line	\$1,342,000	6% (\$80,000)
	Total allocated to project		\$1,889,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.